

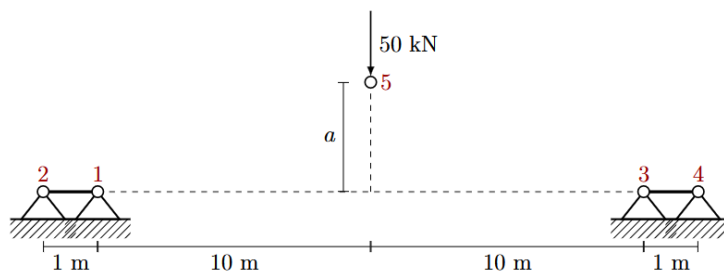


ME-473: DYNAMIC FINITE ELEMENT ANALYSIS OF STRUCTURES

Mini-project 2 report - Design of a 2d truss bridge

GROUP F

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April 18th, 2025

1 Introduction

This report presents the analytical and numerical solutions obtained from the design and analysis of a two dimensional truss bridge. The objective of this project is to create three designs to bridge a twenty meter gap, while a force is applied in the middle. The three bridges will be made of either wood, aluminium, or steel and will need to respect certain constraints such as stress, node displacement, natural frequencies and total volume used. The design will be analyzed using the Finite Element Method to assess the characteristics of each bridge and make sure they follow the corresponding constraints. The software used for this study will be MATLAB.

2 Methodology

2.1 Parameters

The starting steps are four fixed nodes with coordinates:

- Left outer: $(-1, 0)$
- Left inner: $(0, 0)$
- Right inner: $(20, 0)$
- Right outer: $(21, 0)$

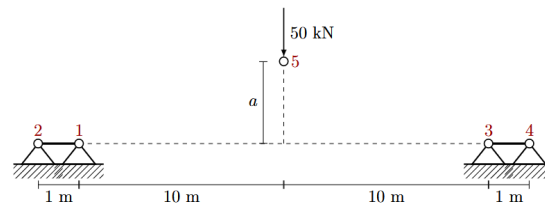


Figure 1: Schematic of the problem setup.

They will be connected through more nodes and trusses to the middle node, where a load of $50kN$ is applied, as shown in Figure 1. This node is positioned at coordinates $(10,a)$, meaning its height can vary freely, but it must remain at the mid-span of the bridge.

The three design will need to be computed for different materials: Steel, Aluminium and Wood, whose properties are summarized in Table 1.

Materials	Properties			
	E (GPa)	Density (kg/m ³)	Cross-Sectional Area (cm ²)	Yield (MPa)
Structural Steel	200	7850	15–50	350 ¹
Aluminium alloy	69	2700	20–70	110 ²
Engineered wood	12	600	30–100	30 ³

Table 1: Summary of material properties.

Finally, the three designs will need to follow certain constraints. The structure must remain within the elastic range, which involves the maximal stress being below the yield strength.

¹Mild Steel from reference [1]

²Aluminium alloy 6061 with T4 temper from reference [2]

³Wood for laminated veneer lumber for structural composite lumber products, Douglas-fir from reference [3]

Moreover, with the current configuration and according to the $L/500$ criterion, the maximal displacement at any node must not exceed 40 mm. The first natural frequency must not fall within the range of 1.5 to 10 Hz and should aim to be above 50 Hz. The total volume of the structure must not exceed the limit:

- Structural steel: 1 m^3
- Aluminium alloy: 1.7 m^3
- Engineered wood: 2 m^3

2.2 Assumptions

Here is a list of all assumptions made:

- All joints are idealized as frictionless pinned connections,
- All properties are homogeneous
- We consider small deformation
- The system must remain in the linear regime
- Trusses experience either compressive or tensile forces
- The gravitational forces are neglected
- Trusses have varying orientations with respect to a fixed global coordinate system, which serves as a stationary reference framework that remains unchanged regardless of the orientation of individual elements

2.3 Numerical simulation

In local coordinates, the equations of motion of a bar are

$$\begin{cases} EA \partial_{x'}^2 u(x', t) = \rho A \ddot{u}(x', t) & x' \in]0, l[\text{ and } t \in]0, T[\\ u(x', 0) = u_0(x') & x' \in]0, l[\\ \dot{u}(x', 0) = v_0(x') & x' \in]0, l[\end{cases}$$

With A being cross-sectional area, E the Young's modulus, ρ the material density, l the length, u the axial displacement and x' the local axial coordinate, according to Figure 2.

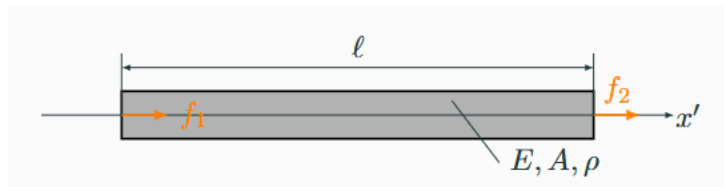


Figure 2: Local diagram of a bar [4].

Each bar will be simulated as a single element using the linear shape functions: $h_1(x') = 1 - \frac{x'}{l}$ and $h_2(x') = \frac{x'}{l}$. The displacement approximation in local coordinates is $u_h(x', t) = h_1(x')q_1(t) + h_2(x')q_2(t)$ and for virtual: $\delta u_h(x') = h_1(x')\delta q_1 + h_2(x')\delta q_2$. From this the local stiffness, mass and load matrices can be computed. With $\frac{\partial h_i}{\partial x'} = h'_i$

$$\mathbf{K}_{\text{loc}} = \int_0^l EA \begin{bmatrix} (h'_1)^2 & h'_1 h'_2 \\ h'_2 h'_1 & (h'_2)^2 \end{bmatrix} dx' = \frac{EA}{l} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\mathbf{M}_{\text{loc}} = \int_0^\ell \rho A \begin{bmatrix} (h_1)^2 & h_1 h_2 \\ h_2 h_1 & (h_2)^2 \end{bmatrix} dx' = \frac{\rho A \ell}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\mathbf{f}_{\text{loc}}(t) = \begin{bmatrix} h_1(0) \\ h_2(0) \end{bmatrix} f_1(t) + \begin{bmatrix} h_1(\ell) \\ h_2(\ell) \end{bmatrix} f_2(t) = \begin{bmatrix} f_1(t) \\ f_2(t) \end{bmatrix}$$

The matrices are then converted from their local frame of reference to the global one through a rotation matrix

$$T = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & \cos(\theta) & \sin(\theta) \end{bmatrix}$$

with θ being the angle between the bar and the reference coordinate system, defined as $\sin(\theta) = \frac{y_2 - y_1}{\ell}$ and $\cos(\theta) = \frac{x_2 - x_1}{\ell}$.

The global matrix, for each element, then become:

$$\mathbf{K} = \mathbf{T}^T \mathbf{K}_{\text{loc}} \mathbf{T} = \frac{EA}{\ell} \begin{bmatrix} \cos^2(\theta) & \sin(\theta) \cos(\theta) & -\cos^2(\theta) & -\sin(\theta) \cos(\theta) \\ \sin(\theta) \cos(\theta) & \sin^2(\theta) & -\sin(\theta) \cos(\theta) & -\sin^2(\theta) \\ -\cos^2(\theta) & -\sin(\theta) \cos(\theta) & \cos^2(\theta) & \sin(\theta) \cos(\theta) \\ -\sin(\theta) \cos(\theta) & -\sin^2(\theta) & \sin(\theta) \cos(\theta) & \sin^2(\theta) \end{bmatrix}$$

$$\mathbf{M} = \mathbf{T}^T \mathbf{M}_{\text{loc}} \mathbf{T} = \frac{\rho A \ell}{6} \begin{bmatrix} 2 \cos^2(\theta) & 2 \sin(\theta) \cos(\theta) & \cos^2(\theta) & \sin(\theta) \cos(\theta) \\ 2 \sin(\theta) \cos(\theta) & 2 \sin^2(\theta) & \sin(\theta) \cos(\theta) & \sin^2(\theta) \\ \cos^2(\theta) & \sin(\theta) \cos(\theta) & 2 \cos^2(\theta) & 2 \sin(\theta) \cos(\theta) \\ \sin(\theta) \cos(\theta) & \sin^2(\theta) & 2 \sin(\theta) \cos(\theta) & 2 \sin^2(\theta) \end{bmatrix}$$

$$f = T^T f_{\text{loc}} = \begin{bmatrix} f_1 \cos(\theta) \\ f_1 \sin(\theta) \\ f_2 \cos(\theta) \\ f_2 \sin(\theta) \end{bmatrix}$$

These are the matrices that will be reduced and assembled by the MATLAB function to get the full matrices of the problem that will be used in the following equations. The static analysis will be done by solving the equation:

$$Kq = f$$

and the modal analysis [5] by solving:

$$\det(K - \omega_n^2 M) = 0$$

where ω_n are the natural angular frequencies of the system. From this result, the eigenvectors, which will represent the mode shapes, can be computed. The normalization used was the predefined from MATLAB, so that the 2-norm of each eigenvector is 1 [6].

3 Results and Discussion

This section will present the result of the analysis of the three bridges.

3.1 Solutions

Three distinct bridge designs were developed for this project, each constructed using the different materials. The resulting configurations are presented in Figure 3.

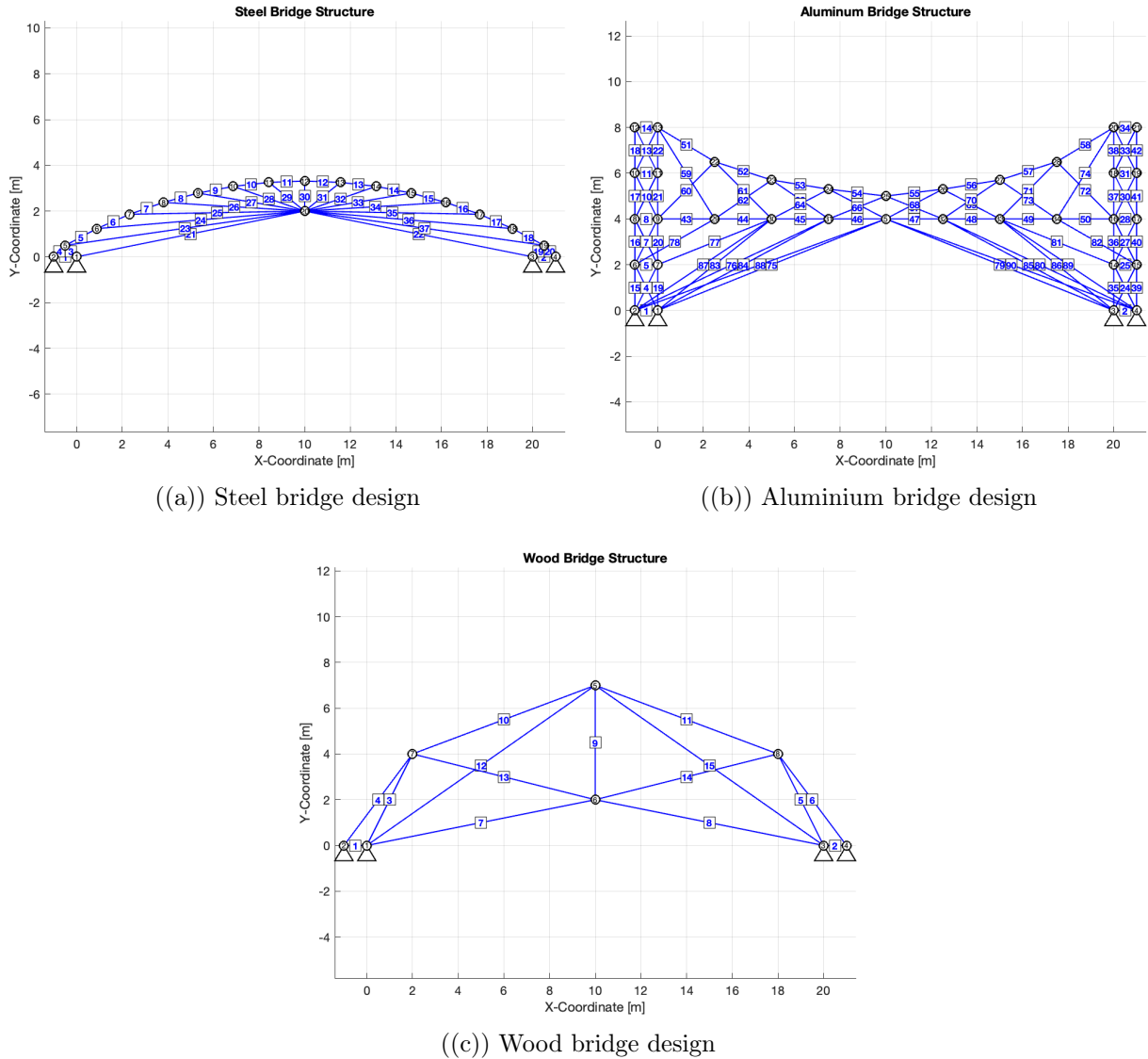


Figure 3: Different bridge designs for each material.

The area chosen for each material was the lowest available, $15mm^2$ for steel, $20mm^2$ for aluminium and $30mm^2$ for wood. Each design features a distinct structural configuration: an arch for steel, a suspended bridge for aluminum, and a revisited dual truss for wood. The choice of structural style was made independently of the material properties and is therefore random.

3.2 Constraints

Table 2 shows the results and performance of the three bridges, with each of them being well below the threshold. It is important to note that the constraint put on frequencies was not to fall in the 1.5-10 Hz range, so the limit is a lower bound unlike the others.

Materials	Stress [MPa]		Displacement[mm]		Frequency[Hz]		Volume [m ³]	
	Actual	Limit	Actual	Limit	Actual	Limit	Actual	Limit
Steel	45.15	350	8.38	40	51.37	10	0.204	1
Aluminium	14.91	110	4.08	40	51.18	10	0.596	1.7
Wood	7.84	30	13.92	40	59.8	10	0.313	2

Table 2: Material performance comparison.

Significant effort was made to ensure that each design had a frequency over 50 Hz and that the total volume was the lowest we could make it. Figure 4 displays the stress distribution and deformation of the bridges under load. The stress in each truss member is visualized using a colour gradient, with the corresponding scale shown on the right. Additionally, nodal displacements are included in the visualization. Nonetheless, due to their small magnitude (less than 14 mm), they are not visually discernible. The point force is also represented on the node with a red arrow.

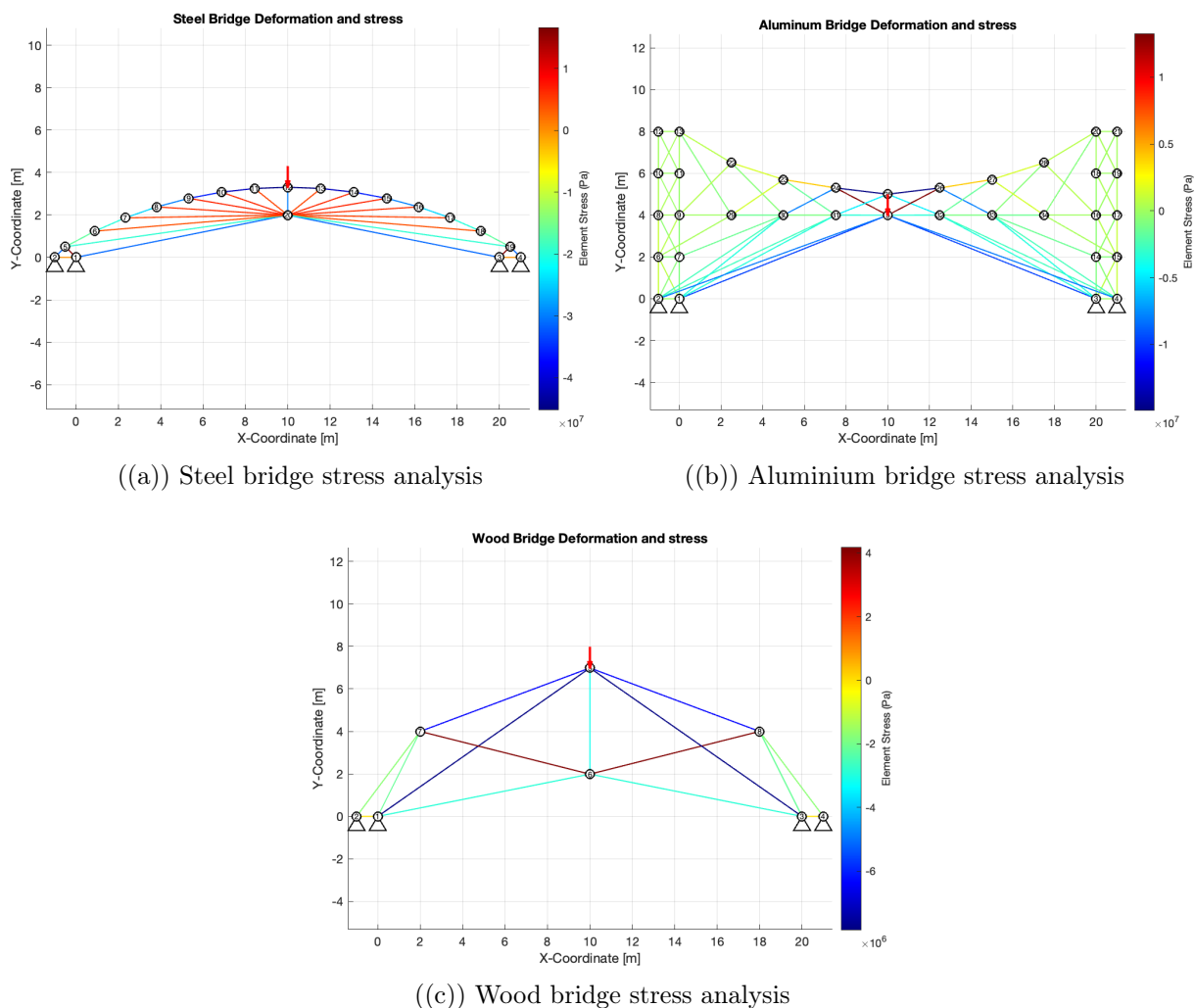


Figure 4: Stress analysis of each bridge design.

3.3 Modes

In this section, the first mode shape of each structure is presented in Figure 5. The first four mode shapes, compared to the undeformed configurations, are available in Appendices A, B, and C. Despite the differences in structural design, the mode shapes appear relatively similar.

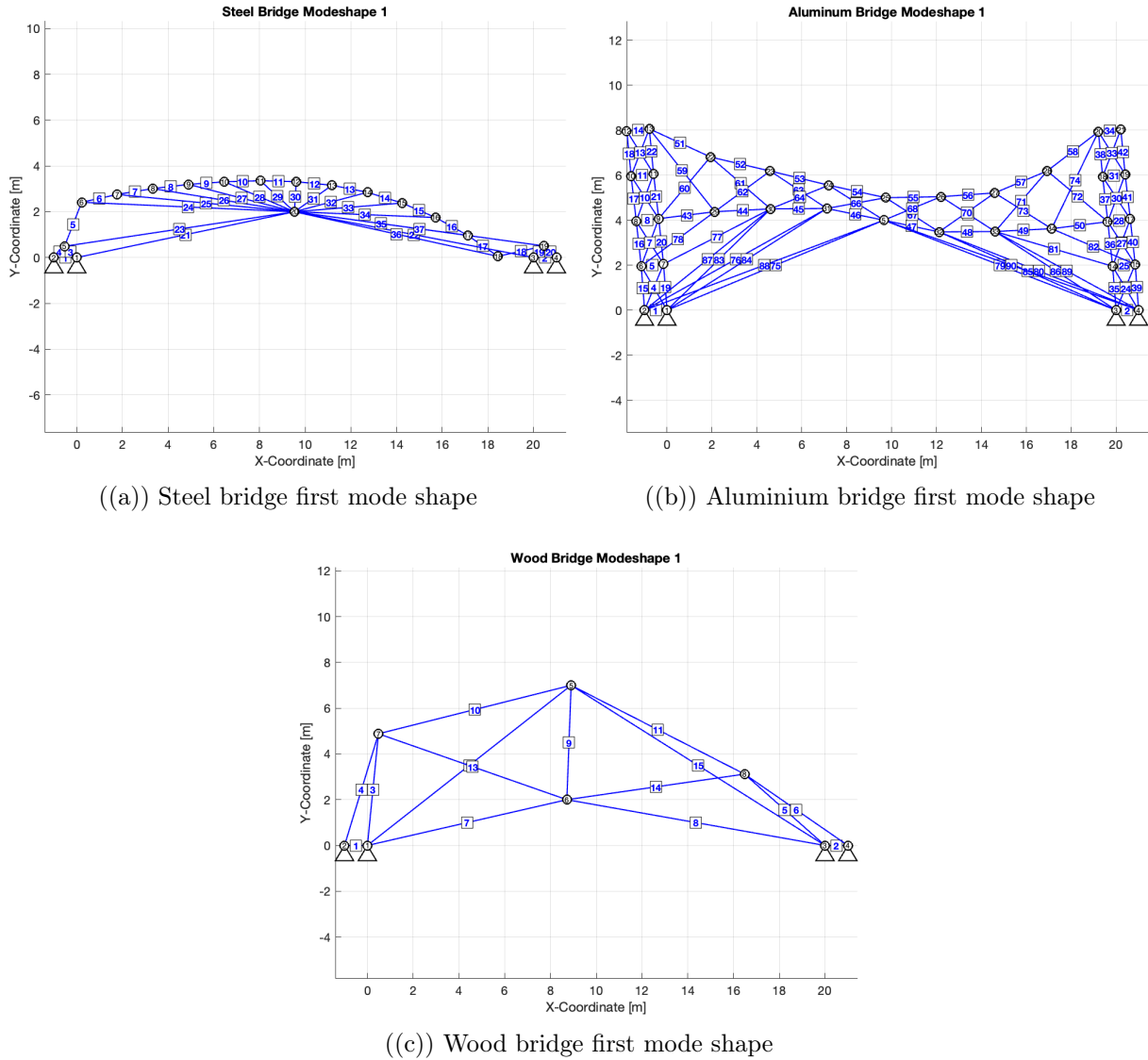


Figure 5: Modal analysis of each bridge design, each mode shape was amplified with a factor 3.

In particular, the steel and wood bridges both exhibit a deformation pattern that combines leftward translation with clockwise rotation, alternating with rightward translation and counter-clockwise rotation — as it is oscillatory. The aluminum bridge shows a similar pattern, though the rotational component is less pronounced. Looking specifically at the "road" segment (elements 43 to 50), the deformation resembles a pinned–pinned beam undergoing a central rotation due to a moment. This observation indicates that the first mode shape corresponds to a global oscillation of the structure, involving both rigid-body translation and rotation of the whole structure.

3.4 Discussion

The three bridge designs we looked at in this study show different approaches to structuring: an arch for steel, a suspension-style bridge for aluminum, and a revisited dual truss design for wood. Despite their differences, they all have one thing in common: truss layouts that channel the forces from the load to the supports, which helps the bridge handle the load better and reduces deformation.

A key observation is the orientation of the truss elements. Trusses are particularly effective in resisting axial forces. However, they do not have a resistance to bending or forces acting in directions perpendicular to their axis. This is clearly demonstrated in Figure 6, where a lateral force applied to the truss results in an infinite displacement. In contrast, forces applied along

the axis of the truss are efficiently resisted, with minimal movement at the nodes. The stiffness matrix will be zero in the direction of the force if it is perpendicular to the truss, and will be maximized when the force is aligned with the truss's axis. Therefore, aligning the trusses with the primary direction of the applied loads improves the overall stiffness of the structure.

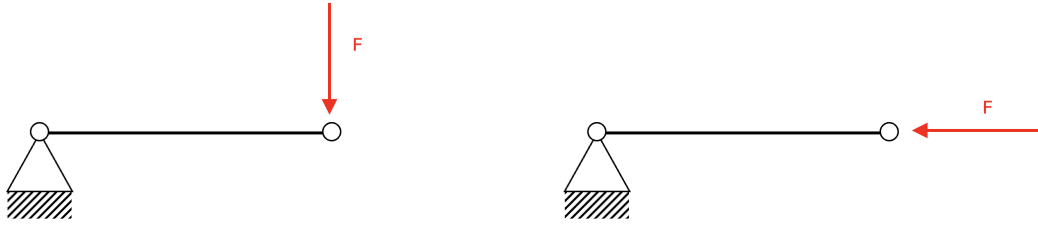


Figure 6: Thought experiment: load comparison on a truss element.

For instance, an aluminum bridge design (code of the geometry in Appendix D) that failed due to an excessive number of horizontal members and insufficient vertical structure resulted in poor load distribution shown in the Figure 7 below. The structure also has a large aspect ratio, with a height of 0.5 m and a length of 20 m. It was composed entirely of small triangular elements, each measuring 0.5 m in both length and height.

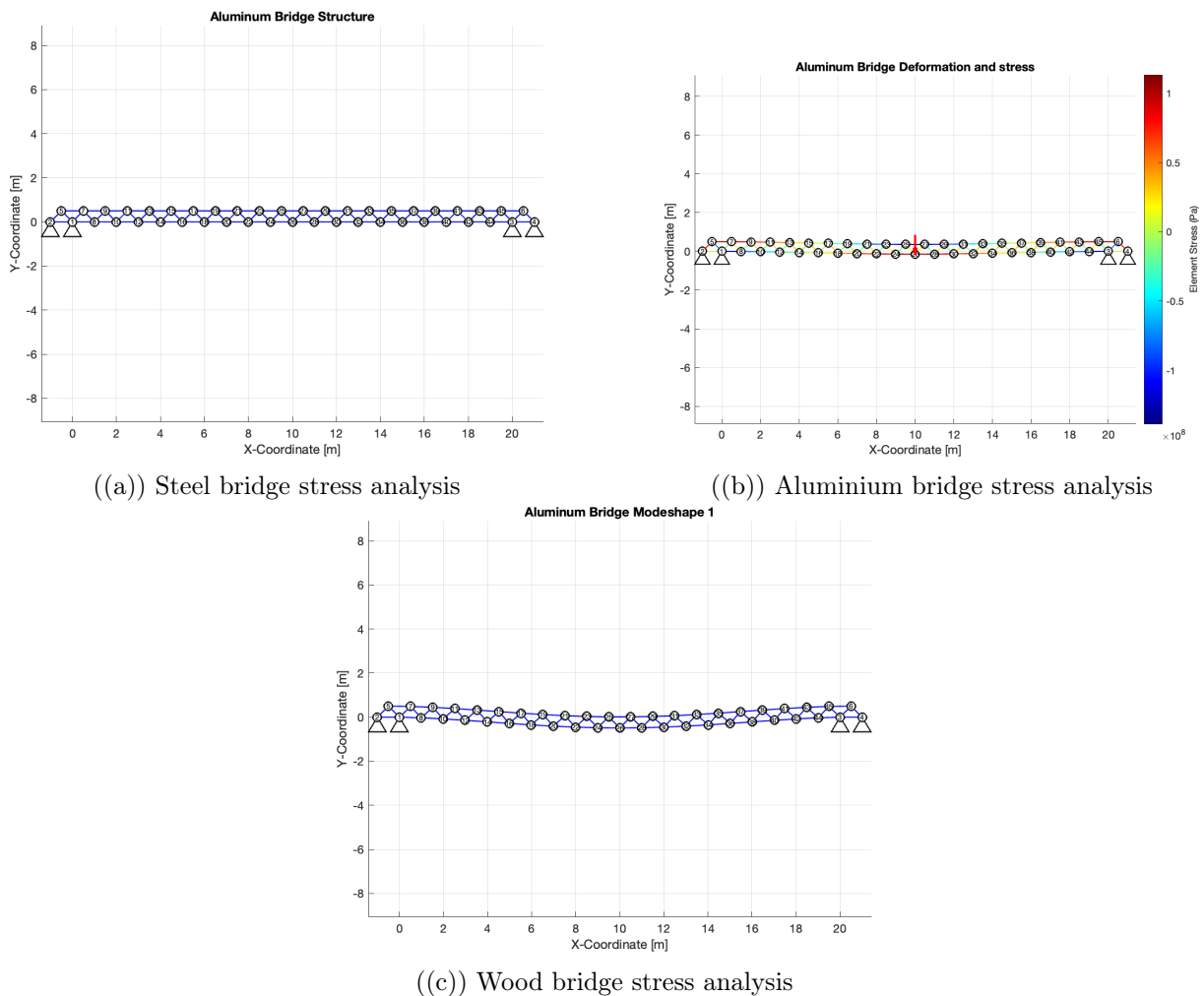


Figure 7: Result plots of a failed flat bridge design.

Several issues contributed to its failure: the stress in the structure exceeded the material's yield limit, reaching 138.46 MPa, compared to the material's limit of 110 MPa and the displacement surpassed the allowable limit by a significant margin, with a deflection of 146.01 mm, far

exceeding the maximum permissible displacement of 40 mm.

These results underline that structural form is as important as material efficiency. Despite its low volume (0.148 m^3), the failed design suffered from a very flat geometry, which limited its ability to redistribute vertical loads effectively. Even with a dense truss layout, the low height meant that most members were oriented horizontally, offering little resistance in the vertical direction. As a result, the structure couldn't guide forces efficiently toward the supports, leading to high stress, large displacements, and poor dynamic performance. This highlights the need for sufficient structural depth to enable effective load transfer. Interestingly, its behaviour closely resembles a clamped-clamped beam.

A comparison between our optimized designs and real-world structures highlights the impact of fundamental engineering strategies on structural performance. Figure 8 presents three iconic bridges: the Golden Gate Bridge, the Pont de la Maltière, and the Viaduc de St-Ursanne. This exemplify the use of curved or triangulated load paths to guide forces efficiently toward their supports. Although differing in scale and function, these structures share a common principle: their geometries are carefully shaped to follow natural force trajectories.



((a)) Golden Gate bridge [7]



((b)) Pont de la Maltière [8]



((c)) Viaduc ferroviaire de St-Ursanne [9]

Figure 8: Bridges around the world.

This design philosophy aligns closely with the successful bridge configurations developed in this study, where optimized truss orientations enhanced stiffness and reduced nodal displacement. The comparison illustrates how theoretical optimization strategies are grounded in real-world engineering practice, reinforcing the importance of geometry.

4 Conclusion

With this report 3 bridge designs were created to explore how a truss structure supports loads and how different material affect that. An analysis was done using the finite element method to verify that each design met the imposed constraints, in terms of volume, frequencies, stress and displacement. The attempt with our structures is to have shapes that redirect the load directly to the anchors. As in the steel shape, where an arc shape was used. These kinds of structure help to redistribute the loads directly to the anchor points.

Further exploration on the subject could be done by restricting the element length, to create a more challenging case, as in video game *Poly Bridge* [10]. In a practical case, we cannot have unlimited length of trusses. Moreover, wood length limit should be shorter than metal, due to manufacturing constraints. E.g. limits of 5m for wood and 8m for metal trusses. This was not taken into consideration with our design. Volume is a good consideration, as it can be related to cost and, if gravity is considered, weight. However, in this study, this constraint was set to high as most of our designs being under 35% of the maximum volume.

Moreover, several simplifications made during the study led to results that may not fully reflect real-world performance. For instance, each truss element was modeled with a small cross-sectional area, which — if assumed to be square or circular — would result in a low second moment of area, making the elements prone to buckling due to their slenderness. A more complex starting set than 2D truss bridge would be more relevant for the given constraints.

References

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A Steel bridge modes

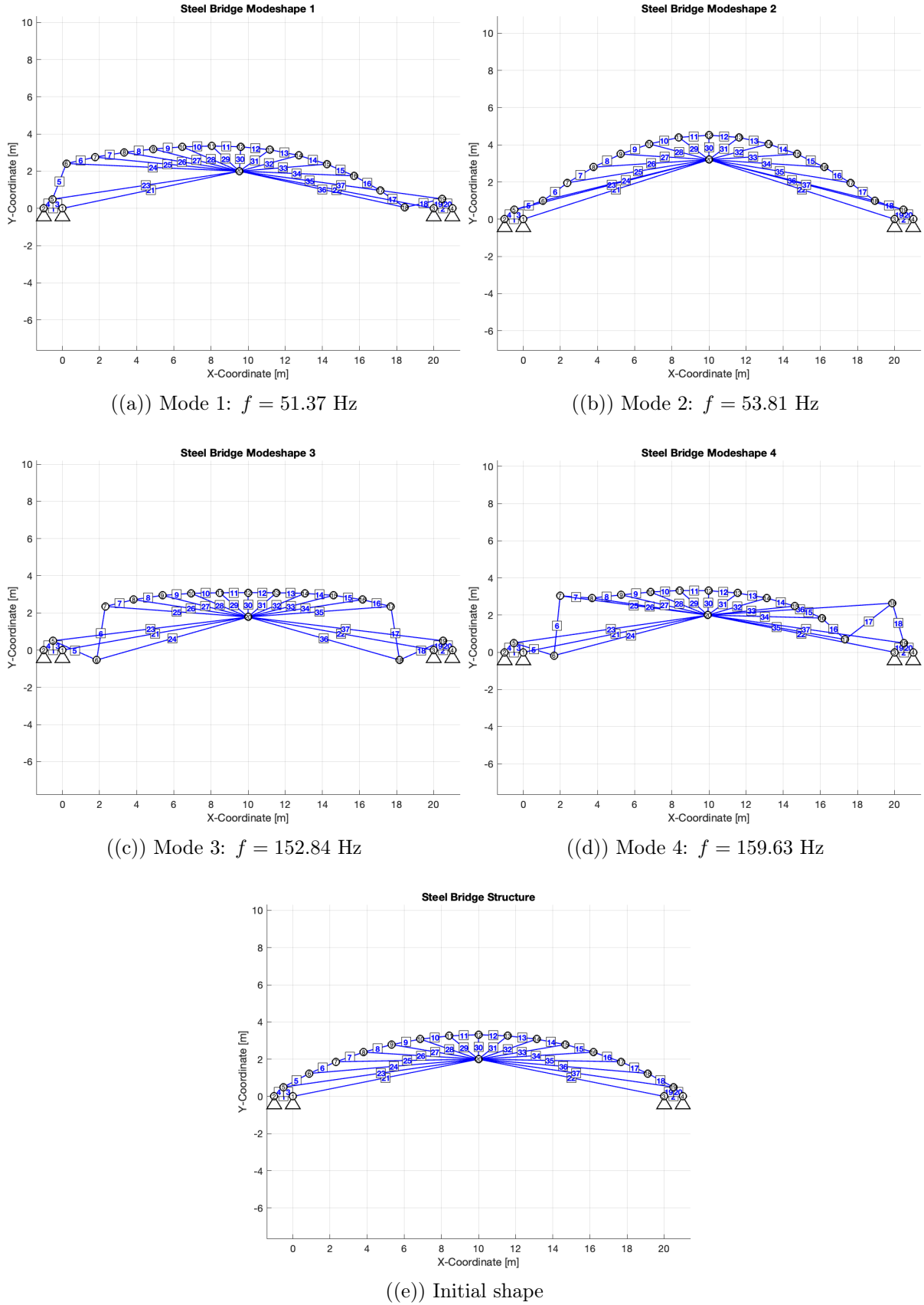


Figure 9: Steel Bridge first 4 mode shapes, a factor 3 was applied to each mode shape, with the initial design to help the comparison.

B Aluminum bridge modes

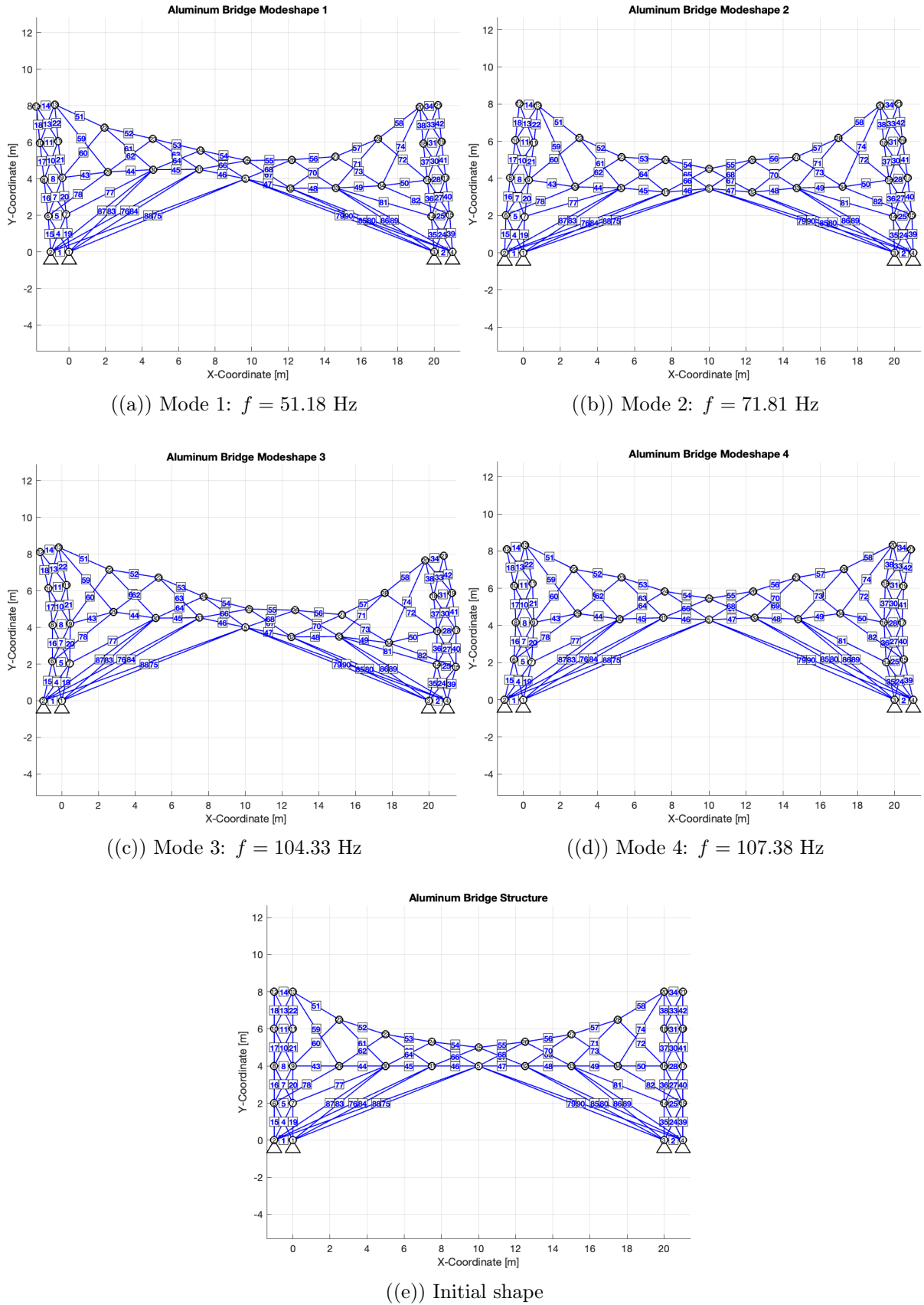


Figure 10: Aluminum Bridge first 4 mode shapes, a factor 3 was applied to each mode shape, with the initial design to help the comparison.

D Aluminum flat bridge with small trusses code

```

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Geometric Construction %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Defining the Nodes location and connectivity table

% Creating a long bridge uniquely made of same size triangles
% Upper nodes of triangle coordinates
x_up = 0.5:1:19.5;
tri_up = [x_up' 0.5*ones(length(x_up),1)];

% Bottom nodes of triangle coordinates
x_down = 1:1:19;
tri_down = [x_down' zeros(length(x_down),1)];

% Creating the nodes coordinate and connectivity table
tri_node = [tri_up(1,:); tri_down(1,:)];
tri_conn = [5 7; 1 7; 1 8; 7 8];

% Deleting the already used nodes 7 and 8
tri_up(1,:) = [];
tri_down(1,:) = [];

% Starting node as node 1 2 3 4 are anchors and 5 6 7 8 cannot be automated
i = 9;

while ((~isempty(tri_up)) || (~isempty(tri_down)))

    if rem(i,2)==0 % if node number is even
        tri_node = [tri_node; tri_down(1,:)];
        tri_conn = [tri_conn; i-2 i; i-1 i];
        tri_down(1,:)=[];
    else % if node number is odd
        tri_node = [tri_node; tri_up(1,:)];
        tri_conn = [tri_conn; i-2 i; i-1 i];
        tri_up(1,:) = [];
    end
    i = i+1;
end

% Concatenating the anchor nodes, nodes 5 and 6 and all the triangles
AluNodes = [anchorCoordinates; -0.5 0.5; 20.5 0.5; tri_node];
AluConnec = [1 2; 3 4; 1 5; 2 5; 3 6; 4 6; tri_conn; ...
            3 size(AluNodes,1)-1; 3 size(AluNodes,1); 6 size(AluNodes,1)];

AluDoF = 2*size(AluNodes,1);

ForceNode = find(AluNodes(:,1)==10);

```